

Alert Triage With Splunk

Using Splunk to investigate alerts, analyze suspicious activity, and improve SOC investigation skills.

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● SCENARIO 1 — Linux Host Brute Force

You've just started your shift as a SOC Analyst at an MSSP. You haven't even gotten around to settling in before the SIEM immediately generates an alert: Brute Force Activity Detection.

1. Initial Alert Analysis

The alert indicates:

- Target Host: tryhackme-2404
- Source IP: 10.10.242.248
- Timestamp: 09:00:21 AM

First key observation:

The source IP appears to be internal, which suggests the attacker may already have access to the corporate network. They may have hacked the VPN, used compromised credentials, or taken over an internal host.

At this stage, there is limited information available about the target host: no asset inventory, no role information. Generic name → likely internal Linux server.

Normal time (09:00), so we can't rely on the timestamp alone to assess it as abnormal.

query 1 — search for failed logins and invalid users

```
index="linux-alert" sourcetype="linux_secure" 10.10.242.248  
| search "Accepted password for" OR "Failed password for" OR "Invalid user"  
| sort + _time
```

I used this initial query to:

- view ALL login attempts (successful + unsuccessful)
- intercept “Invalid user” → a typical account enumeration signal before brute force.

To find failed login attempts on john.smith:

```
index="linux-alert" sourcetype="linux_secure" "Failed password for john.smith"  
| stats count
```

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Nuova ricerca

```
1 index="linux-alert" sourcetype="linux_secure" "Failed password for john.smith"
2 | stats count
```

✓ 500 eventi (prima di 23/11/25 13:25:39,000) Nessun campionamento degli eventi ▼

Eventi Pattern **Statistiche (1)** Visualizzazione

Mostra: 20 per pagina ▼ Formato ▼ Anteprema: on

count ↕

500

To calculate how long the brute-force activity lasted, I ran:

```
index="linux-alert" sourcetype="linux_secure" 10.10.242.248 "Failed password for"
"john.smith"
| stats earliest(_time) as start latest(_time) as end
| eval duration_minutes=round((end-start)/60,0)
| table duration_minutes
```

Press enter or click to view image in full size

Nuova ricerca Salva come ▼

```
1 index="linux-alert" sourcetype="linux_secure" 10.10.242.248 "Failed password for" "john.smith"
2 | stats earliest(_time) as start latest(_time) as end
3 | eval duration_minutes=round((end-start)/60,0)
4 | table duration_minutes
```

✓ 500 eventi (prima di 23/11/25 19:17:43,000) Nessun campionamento degli eventi ▼ Processo ▼ ||| ↗ 🔍

Eventi Pattern **Statistiche (1)** Visualizzazione

Mostra: 20 per pagina ▼ Formato ▼ Anteprema: on

duration_minutes ↕

5

To calculate attack duration for each targeted user account, I used:

```
index="linux-alert" sourcetype="linux_secure" 10.10.242.248 "Failed password for"
| rex field=_raw "Failed password for(?: invalid user)? (?<username>\S+)"
| stats earliest(_time) as start latest(_time) as end count by username
| eval duration_minutes=round((end-start)/60,0)
| table username count duration_minutes
| sort - count
```

Press enter or click to view image in full size

Nuova ricerca

Salva come Crea vista tabella Chiudi

Intervallo temporale: Sempre

```
1 index="linux-alert" sourcetype="linux_secure" 10.10.242.248 "Failed password for"
2 | rex field=_raw "Failed password for?: invalid user"? {username}>{s}"
3 | stats earliest(_time) as start latest(_time) as end count by username
4 | eval duration_minutes=round((end-start)/60,0)
5 | table username count duration_minutes
6 | sort - count
```

508 eventi (prima di 23/11/25 19:21:23,000) Nessun campionamento degli eventi

Processo

Modality intelligente

Eventi Pattern Statistiche (4) Visualizzazione

Mostra: 20 per pagina Formato Antepagina: on

username	count	duration_minutes
john.smith	508	5
david.miller	3	0
sarah.williams	3	0
emma.johnson	2	0

What username was the attacker able to privilege escalate to?

index="linux-alert" sourcetype="linux_secure" john.smith (sudo OR su OR "Successful su" OR "session opened")
| table _time _raw

Press enter or click to view image in full size

Nuova ricerca

Salva come Crea vista tabella Chiudi

Intervallo temporale

```
1 index="linux-alert" sourcetype="linux_secure" john.smith (sudo OR su OR "Successful su" OR "session opened")
2 | table _time _raw
```

10 eventi (prima di 23/11/25 19:25:01,000) Nessun campionamento degli eventi

Processo

Modality intelligente

Eventi Pattern Statistiche (10) Visualizzazione

Mostra: 20 per pagina Formato Antepagina: on

_time	_raw
2025-09-17 09:12:10.762	2025-09-17T09:12:10.762724+00:00 tryhackme-2404 sudo: root : TTY=pts/2 ; PWD=/home/john.smith ; USER=root ; COMMAND=/usr/sbin/adduser system-utm
2025-09-17 09:11:28.984	2025-09-17T09:11:28.984986+00:00 tryhackme-2404 su[3399]: pam_unix(su:session): session opened for user root(uid=0) by john.smith(uid=0)
2025-09-17 09:11:28.979	2025-09-17T09:11:28.979980+00:00 tryhackme-2404 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by john.smith(uid=1001)
2025-09-17 09:11:28.976	2025-09-17T09:11:28.976475+00:00 tryhackme-2404 sudo: john.smith : TTY=pts/1 ; PWD=/home/john.smith ; USER=root ; COMMAND=/usr/bin/su
2025-09-17 09:11:21.181	2025-09-17T09:11:21.181594+00:00 tryhackme-2404 sshd[3327]: pam_unix(sshd:session): session opened for user john.smith(uid=1001) by john.smith(uid=0)
2025-09-17 09:07:25.090	2025-09-17T09:07:25.090742+00:00 tryhackme-2404 (systemd): pam_unix(systemd-user:session): session opened for user john.smith(uid=1001) by john.smith(uid=0)
2025-09-17 09:07:25.043	2025-09-17T09:07:25.043116+00:00 tryhackme-2404 sshd[3106]: pam_unix(sshd:session): session opened for user john.smith(uid=1001) by john.smith(uid=0)
2025-09-17 09:06:01.640	2025-09-17T09:06:01.640839+00:00 tryhackme-2404 (systemd): pam_unix(systemd-user:session): session opened for user john.smith(uid=1001) by john.smith(uid=0)
2025-09-17 09:06:01.594	2025-09-17T09:06:01.594003+00:00 tryhackme-2404 sshd[3007]: pam_unix(sshd:session): session opened for user john.smith(uid=1001) by john.smith(uid=0)
2025-09-17 09:12:10.763	2025-09-17T09:12:10.763758+00:00 tryhackme-2404 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by john.smith(uid=0)

What is the name of the user account created by the attacker for persistence?

index="linux-alert" "useradd"

Press enter or click to view image in full size

Nuova ricerca

Salva come Crea vista tabella Chiudi

Intervallo temporale: Sempre

```
1 index="linux-alert" "useradd"
```

1 evento (prima di 23/11/25 19:29:24,000) Nessun campionamento degli eventi

Processo

Modality intelligente

Eventi (1) Pattern Statistiche Visualizzazione

Formato timeline Zoom indietro Zoom area selezionata Deselezione

1 millisecondo per colonna

Formato Mostra: 20 per pagina Visualizza: Elenco

< Nascondi campi	Tutti i campi	Ora	Evento
CAMPI SELEZIONATI		17/09/25 09:12:10.994	2025-09-17T09:12:10.914743+00:00 tryhackme-2404 useradd[3430]: new user: name=system-utm, UID=1002, GID=1002, home=/home/system-utm, shell=/bin/bash, from /dev/pts/1
host			host = ce-splunk source = auth.log sourcetype = linux_secure

query 2 — number of attempts per user

```

index="linux-alert" sourcetype="linux_secure" 10.10.242.248
| rex field=_raw "^\\d{4}-\\d{2}-\\d{2}T[^\\s]+\\s+(?<log_hostname>\\S+)"
| rex field=_raw "sshd\\[\\d+\\]:\\s*(?<action>Failed|Accepted)\\s+\\S+\\s+for(?: invalid user)?
(?<username>\\S+) from (?<src_ip>\\d{1,3}(?:\\.\\d{1,3}){3})"
| eval process="sshd"
| stats count values(src_ip) as src_ip values(log_hostname) as hostname values(process) as
process by username

```

This query extract username, src_ip and action and then aggregate.

query 3 — Verifying Brute Force Success

The next step was determining whether the attacker successfully gained access:

Did the attacker gain entry?

```

index="linux-alert" sourcetype="linux_secure" 10.10.242.248
| rex field=_raw "^\\d{4}-\\d{2}-\\d{2}T[^\\s]+\\s+(?<log_hostname>\\S+)"
| rex field=_raw "sshd\\[\\d+\\]:\\s*(?<action>Failed|Accepted)\\s+\\S+\\s+for(?: invalid user)?
(?<username>\\S+) from (?<src_ip>\\d{1,3}(?:\\.\\d{1,3}){3})"
| eval process="sshd"
| stats count values(action) values(src_ip) as src_ip values(log_hostname) as hostname
values(process) as process by username

```

● SCENARIO 2 — Windows Scheduled Task Persistence

Alert: Potential Task Scheduler Persistence Identified

- Time: 30/08/2025–10:06:07 AM
- Host: WIN-H015
- User: oliver.thompson
- Task Name: AssessmentTaskOne

Your mission: determine whether this activity is suspicious.

Initial Analysis Before Investigating in Splunk

- Host type
The device name is WIN-H015.
Prefix “WIN-” → typically workstation, not server.
If it were “SRV-”, “MSQL-”, or “WEB-”, it would be a server.

→ So you’re dealing with a user endpoint, not a server.

- Who is the user?
The room says that oliver.thompson is a Systems Engineer.

This is important because:

An IT technician can actually create scheduled tasks.

But... it's not normal to create them at a specific time every day, neither with certutil nor by pointing to strange domains.

→ Even though the user works in IT, the activity still appears suspicious.

- Time

10:06 AM → business hours, so not an out-of-hours alert.

However, this does not eliminate the possibility that the account has been compromised.

Query:

```
index="win-alert" EventCode=4698 AssessmentTaskOne
```

```
| table _time EventCode user_name host Task_Name Message
```

EventCode 4698 = *A scheduled task was created*

Press enter or click to view image in full size

1 index="win-alert" EventCode=4698 AssessmentTaskOne
2 | table _time EventCode user_name host Task_Name Message

✓ 1 evento (prima di 23/11/25 19:54:04,000) Nessun campionamento degli eventi

Processo

Intervallo temporale: Sempre

Eventi Pattern Statistiche (7) Visualizzazione

Mostra: 20 per pagina Formato Antepagina: on

_time	EventCode	user_name	host	Task_Name	Message
2025-08-08 10:06:07	4698	oliver.thompson	WIN-H015	\AssessmentTaskOne	A scheduled task was created.

Subject:

Security ID: S-1-5-21-1566530601-3185518712-10604624-1811
Account Name: oliver.thompson
Account Domain: WIN-H015
Logon ID: 0x7824f

Task Information:

Task Name: \AssessmentTaskOne
Task Content: <?xml version="1.0" encoding="UTF-16"?>
<Task version="1.2" xmlns="http://schemas.microsoft.com/windows/2004/02/mtl/task">
<RegistrationInfo>
<Data>2025-08-08T10:06:07</Data>
<Author>WIN-H015\oliver.thompson</Author>
<URI>\AssessmentTaskOne</URI>
</RegistrationInfo>
<Triggers>
<CalendarTrigger>
<StartBoundary>2025-08-08T10:15:00</StartBoundary>
<Enabled>true</Enabled>

There is only one event associated with that task.

So the change is isolated to a single host and a single user → suspect.

</ TASK >

Other Information:

ProcessCreationTime: 81346268269380621

ClientProcessId: 5816

ParentProcessId: 4128

FQDN: 0

What is the name of the parent process for the process that created this malicious task?

index="win-alert" ProcessId=4128

Press enter or click to view image in full size

Time

Event

8/30/25

9:53:46.000 AM

08/30/2025 09:53:46 AM

LogName=Microsoft-Windows-Sysmon/Operational

EventCode=1

EventType=4

ComputerName=WIN-H015

Show all 38 lines

Event Actions

Type	Field	Value	Actions
Selected	ParentProcessId	5100	
	host	WIN-H015	
	source	WinEventLog:Microsoft-Windows-Sysmon/Operational	
	sourcetype	WinEventLog	
Event	CommandLine	"C:\Windows\system32\cmd.exe"	

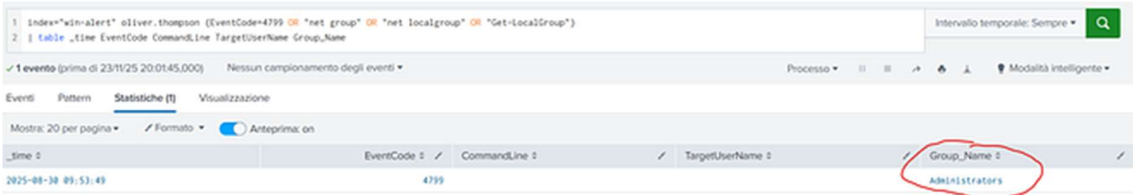
Which local group did the attacker enumerate during discovery?

index="win-alert" oliver.thompson (EventCode=4799 OR "net group" OR "net localgroup" OR "Get-LocalGroup")

| table _time EventCode CommandLine TargetUserName Group_Name

This query was used to identify local group enumeration activity through EventCode 4799 and related commands.

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1 index="win-alert" oliver.thompson (EventCode=4799 OR "net group" OR "net localgroup" OR "Get-LocalGroup") 2 table _time EventCode CommandLine TargetUserName Group_Name					Intervallo temporale: Sempre 🔍
✓ 1 evento (prima di 23/11/25 20:01:45,000) Nessun campionamento degli eventi ▾					
Processo ▾ 🔍 ⬆️ ⬇️ Modaltà intelligente ▾					
Eventi Pattern Statistiche (1) Visualizzazione					
Mostra: 20 per pagina ▾ Formato ▾ Antepriima: on					
_time s	EventCode s	CommandLine s	TargetUserName s	Group_Name s	
2025-08-30 09:53:49	4799			Administrators	

What is the name of the workstation from which the Threat Actor logged into this host?

Search login events (EventCode 4624):

index="win-alert" oliver.thompson EventCode=4624

| table _time Workstation_Name WorkstationName Source_Network_Address LogonType

Press enter or click to view image in full size

1 Index="win-alert" oliver.thompson EventCode=4624					
2 table _time Workstation_Name WorkstationName Source_Network_Address LoginType					
Intervallo temporale: Sempre					
7 eventi (prima di 23/10/25 20:03:37:000) Nessun campionamento degli eventi					
Processo					
Modalità intelligente					
Eventi Pattern Statistiche (7) Visualizzazione					
Mostra: 20 per pagina Formato Anteprima: on					
_time	Workstation_Name	WorkstationName	Source_Network_Address	LoginType	
2025-08-30 09:50:33	WIN-H015		10.10.157.155		
2025-08-30 09:50:33	WIN-H015		10.10.157.155		
2025-08-30 09:50:32	DEV-QA-SERVER		10.10.157.155		
2025-08-30 09:50:30	DEV-QA-SERVER		10.10.157.155		
2025-08-30 09:42:01	WIN-H015		10.14.94.82		
2025-08-30 09:42:01	WIN-H015		10.14.94.82		
2025-08-30 09:41:59	-		10.14.94.82		

SCENARIO 3 — Web Shell Exploitation su web.trywinme.thm

You just received an alert:

Potential Web Shell Upload Detected

Time: 14/09/2025, 09:31:51

Host / Resource: <http://web.trywinme.thm>

Suspicious IP: 171.251.232.40

Before jumping into Splunk, I first reviewed the alert details and assessed the potential severity of the incident.

Preliminary analysis of the alert

Resource

web.trywinme.thm → is the organization's official web server.

Any suspicious activity on a web server is highly critical, because it is public and likely exposes WordPress.

Suspicious IP

171.251.232.40 → Vietnamese IP → immediately check Threat Intel.

Threat intelligence checks showed:

malicious >3000 times

associated with brute force and scanning activities

→ The alert already has a high probability of being a True Positive.

query:

index=web-alert 171.251.232.40

| table _time clientip useragent uri_path method status

| sort + _time

The investigation revealed the following:

User-Agent = Hydra

Hydra is a brute force tool → the attacker is trying to:

- credentials enumeration
- massive attempts on wp-login.php

→ First sign of a password spraying / brute force attack.

Let's remove Hydra and analyze the rest.

```
index=web-alert 171.251.232.40 useragent!="Mozilla/5.0 (Hydra)"
| table _time clientip useragent uri_path referer referer_domain method status
```

And here's the key point:

POST on admin-ajax.php

theme-editor.php?file=b374k.php

This activity immediately appeared suspicious because:

The theme editor should NOT reference arbitrary .php files.

The filename b374k.php is commonly associated with PHP web shells.

→ Here we see the pivot point:

The activity appeared to progress from brute-force attempts to direct interaction with a web shell through the theme editor.

Detailed Web Shell Investigation

```
index=web-alert 171.251.232.40 b374k.php
| table _time clientip useragent uri_path referer method status
| sort + _time
```

What time did the brute-force activity using Hydra begin?

Answer Format Example: 2025-01-15 12:30:45

query:

```
index=web-alert 171.251.232.40 "wp-login.php"
| search useragent="Mozilla/5.0 (Hydra)"
| sort + _time
| table _time clientip uri_path method status useragent
```

Press enter or click to view image in full size

1 index=web-alert 171.251.232.40 "wp-login.php"
 2 | search useragent="Mozilla/5.0 (Hydra)"
 3 | sort + _time
 4 | table _time clientip uri_path method status useragent

✓ 316 eventi (prima di 24/11/25 09:24:11,000) Nessun campionamento degli eventi

Processo

Eventi Pattern **Statistiche (316)** Visualizzazione

Mostra: 20 per pagina Formato Antepagina: on

_time	clientip	uri_path	method	status	useragent
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	GET	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	GET	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	GET	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	GET	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	GET	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	POST	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	POST	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	POST	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	POST	200	Mozilla/5.0 (Hydra)
2025-09-14 21:20:27	171.251.232.40	/wp-login.php	POST	200	Mozilla/5.0 (Hydra)

Which user agent did the attacker use when interacting with the web shell?

query:

index=web-alert 171.251.232.40 b374k.php
 | table _time uri_path method useragent referer status
 | sort + _time

Press enter or click to view image in full size

Nuova ricerca

1 index=web-alert 171.251.232.40 b374k.php
 2 | table _time uri_path method useragent referer status
 3 | sort + _time

✓ 5 eventi (prima di 24/11/25 09:27:00,000) Nessun campionamento degli eventi

Processo

Eventi Pattern **Statistiche (5)** Visualizzazione

Mostra: 20 per pagina Formato Antepagina: on

_time	uri_path	method	useragent	referer	status
2025-09-14 21:22:46	/wp-admin/theme-editor.php	GET	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	http://web.trywinne.thu/wp-admin/theme-editor.php	200
2025-09-14 21:31:51	/wp-admin/admin-ajax.php	POST	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	http://web.trywinne.thu/wp-admin/theme-editor.php?file=b374k.php&theme=blocksky	200
2025-09-14 22:04:13	/wp-admin/admin-ajax.php	POST	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	http://web.trywinne.thu/wp-admin/theme-editor.php?file=b374k.php&theme=blocksky	200
2025-09-14 22:05:17	/wp-admin/admin-ajax.php	POST	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	http://web.trywinne.thu/wp-admin/theme-editor.php?file=b374k.php&theme=blocksky	200
2025-09-14 22:07:18	/wp-admin/admin-ajax.php	POST	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36	http://web.trywinne.thu/wp-admin/theme-editor.php?file=b374k.php&theme=blocksky	200

What was the number of requests made by the attacker to the server via the web shell?

To count all requests to b374k.php the query is:

index=web-alert 171.251.232.40 b374k.php
 | stats count

The initial count included both GET and POST requests, so I filtered specifically for POST requests to identify successful interactions.

After filtering the logs, I confirmed that there were four successful POST requests.

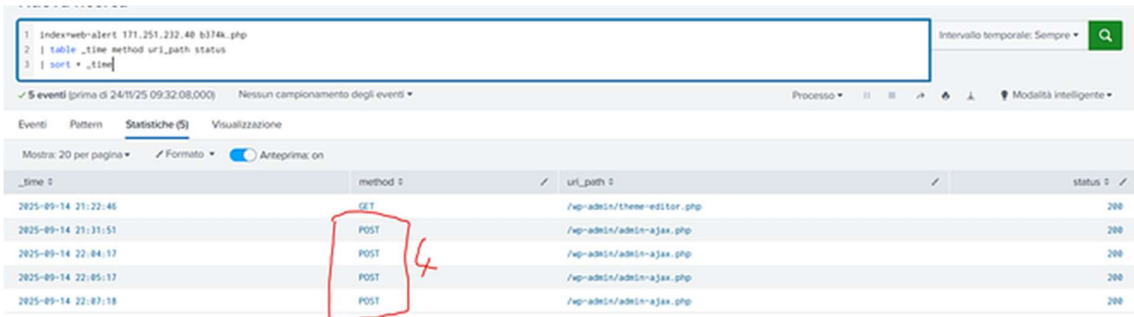
query:

index=web-alert 171.251.232.40 b374k.php

| table _time method uri_path status

| sort + _time

Press enter or click to view image in full size



The screenshot shows a Splunk search interface. At the top, a search bar contains the query: `index=web-alert 171.251.232.40 b374k.php`, `| table _time method uri_path status`, and `| sort + _time`. Below the search bar, the results are displayed in a table. The table has columns for `_time`, `method`, `uri_path`, and `status`. The results show a GET request followed by three POST requests. A red box highlights the three POST requests, and a red handwritten '4' is next to them, indicating the total count of POST requests.

_time	method	uri_path	status
2025-09-14 21:22:46	GET	/wp-admin/theme-editor.php	200
2025-09-14 21:31:51	POST	/wp-admin/admin-ajax.php	200
2025-09-14 22:04:17	POST	/wp-admin/admin-ajax.php	200
2025-09-14 22:05:17	POST	/wp-admin/admin-ajax.php	200
2025-09-14 22:07:18	POST	/wp-admin/admin-ajax.php	200

ans: 4

👉 Written and documented by M. P. Pranav Kumar as part of my hands-on cybersecurity learning journey.